A Systematic Evaluation of Classifiers and Preprocessing for Film Categorization The Tournament

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Curso de Estadística Inferencial

Grid Search Configuration - Tree-Based & KNN Models

Family	Variant	Hyperparameter Ranges
Tree-Based Classifiers		
	DecisionTree	`max_depth`: [3,5,7,10,20] `min_samples_split`: [2,5,10] `criterion`: [gini, entropy]
	RandomForest	`n_estimators`: [50,100,200] `max_depth`: [3,5,7,10] `min_samples_split`: [2,5,10]
	ExtraTrees	`n_estimators`: [50,100,200] `max_depth`: [3,5,7,10,20] `min_samples_split`: [2,5,10]
K-Nearest Neighbors		
	KNN_Coarse	`n_neighbors`: [10,30,50,70,90] `weights`: [uniform, distance]
	KNN_Medium	`n_neighbors`: [10,15,20,25] `weights`: [uniform, distance]
	KNN_Fine	`n_neighbors`: [1,2,3] `weights`: [uniform, distance]
	KNN_Weighted	`n_neighbors`: [5,10,15] `weights`: [distance]
	KNN_Cosine	`n_neighbors`: [5,10,15] `weights`: [uniform, distance] `metric`: [cosine]

Grid Search Configuration - Probabilistic & SVM Models

Family	Variant	Hyperparameter Ranges
Probabilistic Models		
	GaussianNB	var_smoothing: [1e-9, 1e-8, 1e-7, 1e-6]
	LDA	solver: [svd, lsqr, eigen] shrinkage: [None, auto, 0.1, 0.5, 0.9]
Support Vector Machines		
	LinearSVM	C: [0.1, 1, 10, 100] kernel: [linear]
	QuadraticSVM	C: [0.1, 1, 10] gamma: [scale, auto] kernel: [poly], degree: [2]
	CubicSVM	C: [0.1, 1, 10] gamma: [scale, auto] kernel: [poly], degree: [3]
	RbfSVM	C: [0.1, 1, 10, 100] gamma: [scale, auto] kernel: [rbf]

Grid Search Configuration - Logistic Regression & Neural Networks

Family	Variant	Hyperparameter Ranges
Logistic Regression		
	LogisticRegression	`C`: [0.001, 0.01, 0.1, 1, 10] `penalty`: [l2]
	Logistic_L1	`C`: [0.001, 0.01, 0.1, 1, 10] `penalty`: [l1], `solver`: [saga]
	Logistic_EN	`C`: [0.001, 0.01, 0.1, 1, 10] `penalty`: [elasticnet] `l1_ratio`: [0.1, 0.5, 0.9], `solver`: [saga]
Neural Networks		
	Perceptron	`alpha`: [0.0001, 0.001, 0.01] `penalty`: [None, l2, l1, elasticnet]
	MLP_1Layer	`hidden_layer_sizes`: [(50,), (100,), (200,)] `alpha`: [0.0001, 0.001, 0.01] `learning_rate`: [constant, adaptive]
	MLP_2Layer	`hidden_layer_sizes`: [(50,25), (100,50), (200,100)] `alpha`: [0.0001, 0.001, 0.01] `learning_rate`: [constant, adaptive]

Data Preprocessing Pipeline

```
function LOAD_DATA(datasets_path, dataset_name)
    file_path ← Path(datasets_path)/dataset_name
    data ← pd.read_csv(file_path)
    data ← data[data['Time_taken'].notna()]                              ▷ Remove missing
    d3_dummies ← pd.get_dummies(data['3D_available'], prefix='3D_available', dtype = int)
    d3_dummies ← d3_dummies.drop('3D_available_NO', axis = 1)
    data ← pd.concat([data.drop('3D_available', axis = 1), d3_dummies], axis = 1)
    genre_dummies ← pd.get_dummies(data['Genre'], prefix=' Genre', dtype = int)
    genre_dummies ← genre_dummies.drop('Genre_Thriller', axis = 1)
    data ← pd.concat([data.drop('Genre', axis = 1), genre_dummies], axis = 1)
    return data, file_path.stem
end function
```

Input

- Raw CSV
- Categorical vars
- Missing values

Processing

- Filter missing
- One-hot encode
- Drop ref cats

Output

- Clean data
- Encoded features
- Filename stem

My Experiment's Results - Part 1/3

Model	Dataset	F1 Score
RandomForest	shuffle_01	0.602 564
	shuffle_02	0.645 963
	shuffle_03	0.681 081
	shuffle_04	0.654 545
	shuffle_05	0.691 358
	shuffle_06	0.666 667
	shuffle_07	0.651 163
	shuffle_08	0.666 667
	shuffle_09	0.603 774
	shuffle_10	0.670 732
KNN_Medium	shuffle_01	0.589 744
	shuffle_02	0.612 245
	shuffle_03	0.633 540
	shuffle_04	0.513 158
	shuffle_05	0.637 500
	shuffle_06	0.584 416
	shuffle_07	0.629 630
	shuffle_08	0.589 744
	shuffle_09	0.513 889
	shuffle_10	0.543 046

Table: My Experiment's Results - RandomForest and KNN_Medium Models

My Experiment's Results - Part 2/3

Model	Dataset	F1 Score
LDA	shuffle_01	0.641 509
	shuffle_02	0.566 038
	shuffle_03	0.658 537
	shuffle_04	0.696 203
	shuffle_05	0.619 355
	shuffle_06	0.632 911
	shuffle_07	0.658 065
	shuffle_08	0.616 352
	shuffle_09	0.620 253
	shuffle_10	0.666 667
LinearSVM	shuffle_01	0.634 146
	shuffle_02	0.606 452
	shuffle_03	0.693 642
	shuffle_04	0.645 161
	shuffle_05	0.591 195
	shuffle_06	0.632 258
	shuffle_07	0.675 159
	shuffle_08	0.609 756
	shuffle_09	0.578 616
	shuffle_10	0.685 714

Table: My Experiment's Results - LDA and LinearSVM Models

My Experiment's Results - Part 3/3

Model	Dataset	F1 Score
LogisticRegression	shuffle_01	0.600 000
	shuffle_02	0.612 500
	shuffle_03	0.682 353
	shuffle_04	0.700 000
	shuffle_05	0.592 105
	shuffle_06	0.627 451
	shuffle_07	0.675 159
	shuffle_08	0.625 000
	shuffle_09	0.629 630
	shuffle_10	0.678 363
MLP_2Layer	shuffle_01	0.587 500
	shuffle_02	0.584 416
	shuffle_03	0.593 023
	shuffle_04	0.670 807
	shuffle_05	0.602 410
	shuffle_06	0.583 851
	shuffle_07	0.638 037
	shuffle_08	0.603 774
	shuffle_09	0.632 258
	shuffle_10	0.578 313

Table: My Experiment's Results - LogisticRegression and MLP_2Layer Models

Peer Experiment's Results - Part 1/2

Model	Dataset	F1 Score
RbfSVM	shuffle_01	0.706 383
	shuffle_02	0.656 566
	shuffle_03	0.693 694
	shuffle_04	0.706 383
	shuffle_05	0.706 383
	shuffle_06	0.706 383
	shuffle_07	0.705 645
	shuffle_08	0.696 833
	shuffle_09	0.706 383
	shuffle_10	0.706 383
Logreg	shuffle_01	0.654 320
	shuffle_02	0.677 966
	shuffle_03	0.686 047
	shuffle_04	0.709 677
	shuffle_05	0.658 960

Table: Peer Experiment's Results - RbfSVM and Logreg (Part 1)

Peer Experiment's Results - Part 2/2

Model	Dataset	F1 Score
Logreg	shuffle_06	0.707 182
	shuffle_07	0.630 952
	shuffle_08	0.688 525
	shuffle_09	0.658 537
	shuffle_10	0.681 564
Rf	shuffle_01	0.649 682
	shuffle_02	0.681 564
	shuffle_03	0.609 195
	shuffle_04	0.622 754
	shuffle_05	0.711 864
	shuffle_06	0.666 667
	shuffle_07	0.710 059
	shuffle_08	0.712 644
	shuffle_09	0.674 556
	shuffle_10	0.686 391

Table: Peer Experiment's Results - Logreg (Part 2) and Rf Models

Normality Test Results - Shapiro-Wilk Test

Sample	Statistic (W)	P-value
Sample 1	0.915 250	0.319 032
Sample 2	0.887 070	0.157 131

Table: Shapiro-Wilk normality test results. Null hypothesis (H): Data follows a normal distribution.

Alpha Grid Search Results

Index	Alpha (α)	Beta (β)	Expected Losses
1	0.01	0.964	0.487
2	0.05	0.856	0.453
3	0.10	0.788	0.444
4	0.15	0.688	0.419
5	0.20	0.614	0.407
6	0.25	0.572	0.411
7	0.30	0.512	0.406
8	0.35	0.456	0.403
9	0.40	0.400	0.400
10	0.45	0.338	0.394
11	0.50	0.298	0.399
12	0.55	0.258	0.404
13	0.60	0.218	0.409
14	0.65	0.180	0.415
15	0.70	0.142	0.421
16	0.75	0.108	0.429
17	0.80	0.088	0.444
18	0.85	0.064	0.457
19	0.90	0.036	0.468
20	0.95	0.018	0.484

Table: Alpha Grid Search Results for T-test Coefficients. The optimal confidence value resulted $\alpha = 0.45$.

α	$=$	0.45
β	$=$	0.338
t	$=$	1.285
p_{value}	$=$	0.2157